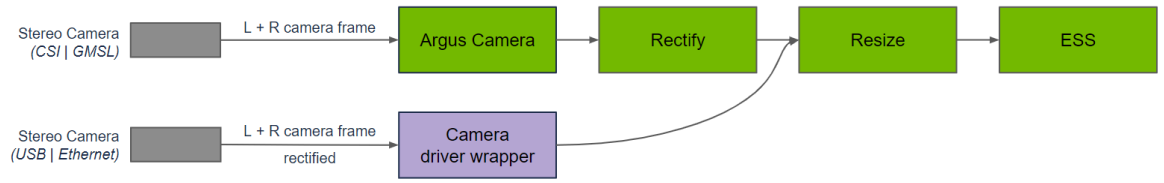
Isaac ROS DNN Stereo Depth

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- Overview
- Quick Experience

Isaac ROS DNN Stereo Depth official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_dnn_stereo_depth/index.html

Overview

DNN Stereo Depth uses deep learning to estimate disparity/depth from stereo images, and can provide distance information for robot manipulation, navigation, and obstacle avoidance. ESS is also the stereo depth component in the Isaac ROS perception solution.



Quick Experience

First use the command below to start the container

```
./run_isaac.sh
```

Then confirm the container on the host with `docker ps`, and open container terminals with `docker exec -it <container name> bash`; each terminal should use the same ROS/FastDDS environment, for example:

```
Jetson Orin Nano L4T R39.2.1 · (8GB + 8GB ZRAM)
RDP 192.168.2.61:3389 jetson/yahboom [LAN]
VNC 192.168.2.61:5900 pass: yahboom [LAN]
jetson@jetson:~$ docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED
      STATUS          PORTS     NAMES
8f397e4b6c3e   my-isaac-ros-slam:4.6-all   "/bin/bash -c '\ncat …"   42 seconds
ago   Up 41 seconds             magical_bhabha
jetson@jetson:~$
```

Note: The terminals below all execute inside the container:

> After the power mode is set to MAXN, an overcurrent warning may pop up when running the case; ignore it.

Terminal A: Start Nano pool=8 ESS, specifying the model size that matches the 1920×1080 input:

```
ros2 launch /workspaces/isaac_ros-dev/launch/ess_nano_pool8.launch.py \
  engine_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.
1.0_onnx_trt10.16/ess.engine \
  ess_plugin_file_path:=/workspaces/isaac_ros-
dev/isaac_ros_assets/models_pending/dnn_stereo_disparity/dnn_stereo_disparity_v4.
1.0_onnx_trt10.16/plugins/aarch64/ess_plugins.so \
  input_image_width:=1920 input_image_height:=1080 \
  model_input_width:=960 model_input_height:=576 threshold:=0.0
```

Terminal B: Play the ESS input bag in a loop:

```
ros2 bag play /workspaces/isaac_ros-
dev/isaac_ros_assets/isaac_ros_ess/rosbags/ess_rosbag \
  --loop --rate 0.25 --remap \
  /left/camera_info:=/left/camera_info_rect \
  /right/camera_info:=/right/camera_info_rect
```

Terminal C: Start the official DNN Stereo visualizer:

```
ros2 run isaac_ros_dnn_stereo_decoder isaac_ros_dnn_stereo_visualizer.py
```



When threshold is set to 0.4, the result is displayed as follows:



Terminal D: Check the disparity output for a short time:

```
timeout 8 ros2 topic hz /disparity
```